



PARTICIPANT'S GUIDE

2020-2021 EDITION

Logistics Game

Powered by **ERPsim**
Compatible with **SAP™ S/4HANA**

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Serious games to learn
enterprise systems and
business analytics

<http://erpsim.hec.ca>

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CHAPTER 1

Introduction

Teaching the concepts underlying an Enterprise Resource Planning (ERP) system is a difficult task. Many students have very little IT experience to which they can relate these concepts. They may have acquired business experience in one or two functional areas, but many of them have only a limited understanding of the operational aspects supporting the value creation process in modern firms. Moreover, they usually have had no firsthand experience with the functional non-integrated software that the ERP system was designed to replace. For these students, the horizontal integration of the firm, one of the greatest benefits of implementing an ERP system, can be very abstract due to their lack of hands-on experience with legacy systems.

Yet business students are very computer-literate these days. Born after the first personal computers came onto the market, many of them have never experienced life without a keyboard or a mouse. Therefore, if they get hands-on experience with an ERP system, undergraduate and graduate students can learn the system and its core concepts very quickly.

The ERP Simulation game is an innovative “learning-by-doing” approach to teaching ERP concepts. During this game, students have to run a business with a real-life ERP (SAP®). Groups of four to five students each operate a firm in a reselling (logistics) context and must interact with suppliers and customers by sending orders, transferring products to regions, and completing the whole end-to-end business cycle. A simulation software program automates the sales process such that each firm receives a large number of orders every round of the simulation. Several administrative functions in SAP® are automated too, so that students can focus on making business decisions, rather than mastery of the many transactions of a complex ERP system. Using a mix of the ERP system’s standard transactions and customized reports, students must analyze information and make business decisions to ensure the profitability of their operations. The learning objectives of this game are fivefold: (i) to develop a hands-on understanding of the concepts underlying enterprise systems, (ii) to experience the benefits of enterprise integration firsthand, (iii) to develop technical skills using ERP software, (iv) to learn how to work in a team, and (v) to learn how to develop, execute and refine strategy in a real-time business environment.

The logistics game has 3 scenarios: Introduction, Extended and Platinum. In this document, we cover the first 2 scenarios.

The Logistics Dairy introductory scenario focuses on the stock transfer logistics and sales processes. Therefore, teams only have to focus on selling their stock of finished products. In the Extended scenario, planning and procurement processes are added, with teams adjusting their sales forecasts, and buying new products to keep on selling. The Logistics Dairy extended game focuses on the integration of all the business processes of the company.

The platinum scenario is similar to the extended game. The difference is that each team member has access to only part of the system that is assigned to their role. Only in round 3 of the game, all team members have access to all parts of the system and can share an integrated system with their teammates.

In [Part I](#) of this book, we introduce the business context and business processes of the logistics scenarios. We will not discuss these with direct reference to the SAP system. It is meant to be a conceptual overview.

In [Part II](#), we cover using the SAP software – how it is used to support the business decisions and operations of our fictitious firm. These include all the transactions covering the full business cycle, whether they are automated in the simulation or performed by students. The section highlights how all the different elements of the business processes are integrated together into the system.



PART 1

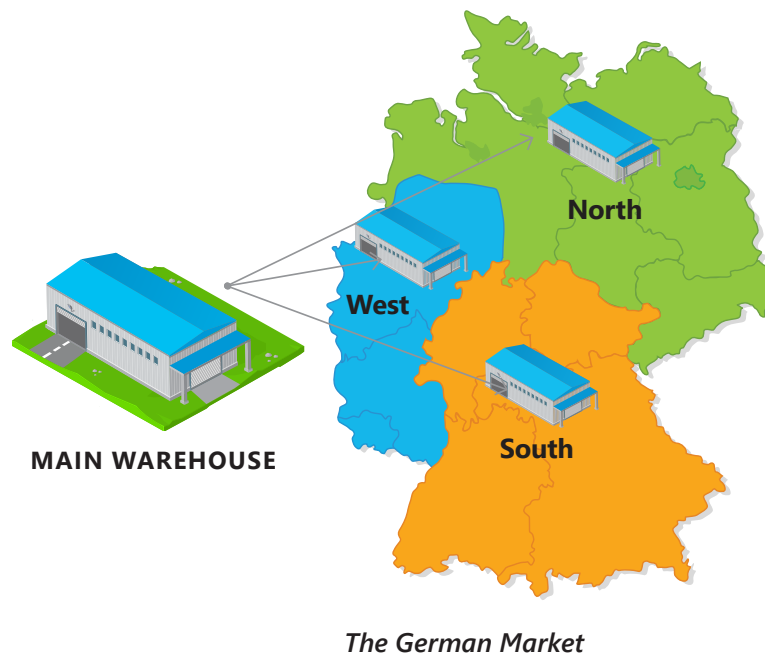
Welcome to Your New Job

As a participant in the HEC Montréal ERP Simulation game, you've just accepted a new job in the dairy industry at the ERPsim Dairy Company. Your new company specializes in reselling dairy products. Over the years, your company has built strong partnerships with multiple dairy product suppliers and sellers around Germany. There are up to 26 local companies competing in this relatively small but competitive market.

Your company's network is now well-established, and your job requires making logistics and business decisions in order to maximize profit. Your company only resells previously acquired products without making any changes. Therefore, reducing logistics costs and achieving operational excellence is the key to success.

All acquired products are delivered to your company's main warehouse based in your central plant. This main warehouse is used as an inspection and sorting center. Every time, you order products from your suppliers, you have to pay a fixed transportation cost.

The German market (figure below) is segmented in three regions (north, west, and south) and your company owns storage locations in each region. Once delivered by your suppliers to the main warehouse, you will have to ship your products to your three storage locations in order to make them available for sale. Again, there are transportation fees for every stock movement you make between your main warehouse and your storage locations. Determining the optimal distribution logistics is the key to maximizing profit and winning the game.



CHAPTER 2

Description of the Marketplace

YOUR PRODUCT DESCRIPTION

Your new company specializes in reselling dairy products. Currently, your company is selling 6 product types: milk, cream, yoghurt, cheese, butter and ice cream.



Logistics Product Types

The table below indicates the description and prices for each of the six products. Highly competitive markets regulate all supplied products; therefore, you cannot negotiate prices with your suppliers.

Logistics Products and their Cost

PRODUCT CODE	DESCRIPTION	UNIT COST
T01	Milk	€ 22.95
T02	Cream	€ 72.07
T03	Yoghurt	€ 25.85
T04	Cheese	€ 82.68
T05	Butter	€ 59.88
T06	Ice Cream	€ 43.15

THE MARKETPLACE

Retail

There is only one distribution channel for dairy products in the local market: traditional retailer stores regrouped into the distribution channel 16. Retail stores offer customers more variety than other types of store. A typical store carries 4 different dairy product brands on their shelves. There are 12 retail stores (4 per area) in the German market. Retail stores are reliable and will pay their bill in 4 days.

Consumers

Data shows that consumers spent an average of €12,000 on dairy products per company, per day (i.e., market size). It is also important to note that consumer preference does not change during the game. Therefore, your goal is to discover the preference in each region based on the information available in reports.

Banks

It may not come as a surprise to you, but your company needs money to make money. You have to purchase and hold inventory ahead of demand. To pay for products you will need cash. Your company has a good relationship with the Weizen Bank. The Bank currently offers you a good credit rate and extends a sufficiently large credit line. After each round, your company will have to produce a full financial statement for your banker.

Your shareholders have invested €500,000. This includes €250,000 invested in assets. This leaves €250,000 in cash. Your responsibility is to effectively manage this money and obtain the highest possible return for your shareholders. The payment of the loan interest is automated, so you will not have to do it manually. The interest is accrued on a daily basis and posted every 5 days. However, be aware that your firm must have a positive cash balance at all times. If your cash account becomes negative, your bank will lend you the required money, however, it will decrease your credit rating and your interest will go up (which negatively affects your company valuation) and you will have to pay far more interest for the rest of the game. Therefore, managing your cash flow effectively during the game will make a big difference.

BUSINESS PROCESSES

Your dairy company, ERPsim Dairy, can only sell products that are held in your regional storage locations. Therefore, (1) you have to buy your products from your suppliers and (2) you need to distribute your product from your main warehouse to your three regional storage locations to be able to sell them. This determines some of the business processes that your company will need to perform.

Because this is a small company, your team will have to supervise almost all of the activities and take an active part in some of the day-to-day processes. There are tasks that you simply cannot delegate, and the success of your company depends on your team. Your responsibilities could include some of the following:

Activities Involved in Integrated Business Processes

ACTIVITIES	DESCRIPTION
Forecasting	All storage locations in the simulation are following a back-to-back trading business processes. Your team needs to forecast demand (i.e., create independent requirements).
Purchase Order Creation	When a purchase requisition is received, the purchasing manager contacts the required product suppliers. After assigning the chosen vendor to the purchase requisition, a formal purchase order is created.
Goods Receipt	When the goods are received, the receiving clerk checks that the goods delivered correspond to the purchase order and records the goods receipt in the system.
Invoice Receipt	The invoice sent with the product is forwarded to accounting and posted by an accounting clerk.
Payment	Since the goods are payable upon an agreed term of payment, the accounting clerk carries out the transaction required to pay the vendor. A check for the amount owed is issued and sent to the vendor.
Stock Transfer Posting	When goods are in the main warehouse, the stock manager needs to determine the level of stock transfer from the main warehouse to the storage locations. After choosing a transfer quantity and the transfer storage location, a formal stock transfer posting is performed. Transportation fees will be charged for each stock transfer. Push or pull methods can be selected.
Sales Order Creation	Each day, several retailers will call your firm to confirm pricing and availability. The firm whose products best meet the needs of each store will secure the sale. If it is your firm, then one of your sales representatives will record a sales order.

ACTIVITIES	DESCRIPTION
Shipping	The products are shipped to the customers and the inventory of products is adjusted (goods issue).
Invoicing	Accounting sends an invoice to the customer.
Incoming Payment Reception	Upon receipt of payment from the customer, the accounting clerk clears the customer account and records the deposit of the payment.

More importantly, there is a list of managerial and strategic activities that you will have to undertake. The strategic decisions involve setting the price and choosing your company's allocation logistics from your main warehouse to your 3 storage locations, which will generate transportation fees. You will also have to manage your cash flow. It is hard and challenging work, but this is why you accepted the job.

MANAGEMENT RESPONSIBILITIES

This section reviews some of the key decisions that your executive team will have to make, depending on which version of the game is being played.

INTRODUCTION SCENARIO

In the Logistics Introduction Game, you will start with initial inventory and the procurement or replenishment process is automated. Therefore, you can focus on decisions regarding selling your products: 1) regional product transfer, and 2) pricing. In the extended game, however, you start with no stock and the procurement process will not be automated.

Procurement (Automatic Replenishment)

The automatic replenishment schedule for the introductory game is as following:

AUTOMATED INVENTORY REPLENISHMENT		
PRODUCT CODE	DESCRIPTION	UNITS (every 5 days)
\$\$-T01	Milk	950
\$\$-T02	Cream	300
\$\$-T03	Yoghurt	700
\$\$-T04	Cheese	350
\$\$-T05	Butter	400
\$\$-T06	Ice Cream	300

This table shows that you start the game with 950 units of Milk, 300 units of Cream and so on. Also, every 5 days, the same amounts will be added to your inventory for all the 6 products, regardless of your sales level or your inventory level.

Since you will have no control over this automatic replenishment, to avoid exceeding your warehousing capacity of 4 000 units in all storage locations, you need to focus on selling your products (via right transfer strategy and right pricing) to avoid stock pile ups.

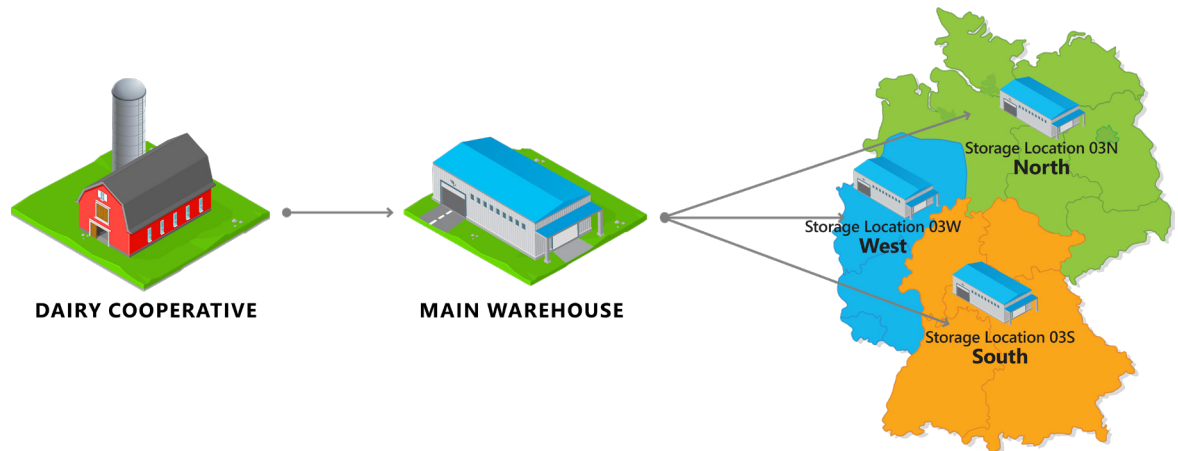
If, however, at some points your sales levels are not enough, and you exceed the warehousing capacity limit, you can rent additional space on a daily basis. The additional space can be rented at a cost of: €50/day for each 1 000 additional units (this is will be €300/day in the extended game).

WAREHOUSING COSTS	
Current Space (Units) (maximum capacity without additional fees)	4 000
Daily Cost per Additional 1 000 Units	€50

Warehousing Cost for the Introductory Game

Regional Product Transfers

One of the most important decisions during the game lies in choosing how much of each product to send from the central (main) warehouse to the regional warehouses. This decision significantly impacts your company's ability to respond to market demand.



The system allows you to choose from two business rules: the push or the pull transfer rule.

Push

Under the push stock transfer rule, you must set the amounts to be transferred for each of the six products and for each of the three storage locations. You must also determine how often (in virtual days) those products must be transferred. Once you have set up these quantities, the simulator will deliver the required quantity to your three storage locations. The simulator will perform the transfer periodically, as specified by the business rule.

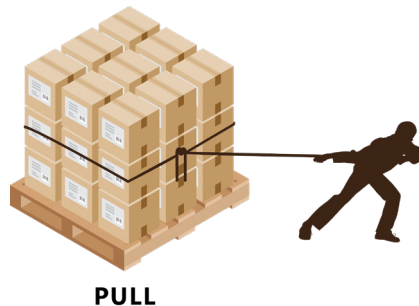
Whenever, there is sufficient stock in your main warehouse to meet the required transfer, the excess material will remain in the central warehouse. Whenever, there is not enough material available in your central warehouse, the generated stock transfers will be proportional to the desired transfers. For example, if only half of the required stock is available, then each location would receive only half of the requested transfer amount.



PUSH

Pull

Under the pull stock transfer rule, you must set regional inventory targets for each of the six products and for each of the three storage locations. You must also determine how often (in virtual days) those products must be transferred. The simulator will calculate, periodically, the difference between the targets and the current quantities in stocks and will transfer the difference. For example, if the target is for 250 units of product 3 in the North, and there are currently 130 units, the simulator tries to transfer only 120 units.



Again, if there is sufficient stock in the central warehouse to meet the calculated transfer, the excess material will remain in the central warehouse. If there is not enough material available in your central warehouse, the stock transfers will be proportional to the calculated transfers.

Hence, three key decisions must be made by your executive team with regards to stock transfer strategy:

- 1) Which business rule (pull or push) should be used?
- 2) How often should the stock transfer be performed? (frequency)
- 3) How many units for each product to each storage locations should be either transferred or set as inventory targets? (quantity)

Two key considerations must be taken into account in order to set your transfer strategy:

- 1) Each time there is a stock transfer from your main warehouse to your storage locations, transportation fees will be charged. Goods allocation is a key element of your strategy, as it will affect your company operating costs as well as allowing quick response to market demand. The transfer fees to each regional storage location are €100.
- 2) Once a unit is sent to one regional warehouse, it cannot be reallocated to another. It will remain in the regional warehouse until it is sold.

If you sent a larger quantity of a product in a given location and the product does not sell, it freezes up valuable storage space. If you transfer small amounts each time and keep sufficient amounts in the central warehouse, you maintain a large level of flexibility. However, it may force you to transfer products more often which generate extra transportation costs. There is, again, a trade-off between higher flexibility and lower logistics costs. In order to set an optimal logistics strategy, it is important to have accurate demand forecast for every product in every region.

Pricing

You may choose to change the selling price for each of your six products whenever you desire. Pricing is a key element of your strategy. If your prices are too high, demand will be low and inventory level will slowly decrease; if your prices are too low, you will not make enough money to cover your expenses.

Optimal pricing depends on many things: (i) the cost of the product; (ii) the transportation and transfer cost; and (iii) demand elasticity. Business acumen is required to make correct pricing decisions. Remember that consumers have other options. They can buy something else if your dairy products are out of their price range. Remember that you also have transportation and logistics costs in addition to acquisition costs; you must price your products correctly so that you do not lose money. As your boss often says, "We aren't running a charity here." You must make a profit.

EXTENDED SCENARIO

Planning and Procurement

As discussed before, in the extended game, you start with no stock (no initial inventory) and the procurement (replenishment) process will not be automated. This means that as a team you will be performing planning and procurement. Therefore, there are 3 processes to be performed in the extended game: 1) planning and procurement, 2) regional product transfer, and 3) pricing.

Since transfer and pricing processes are similar to the introduction game, they won't be repeated here.

As you can only sell products that are in inventory, it is important to make sure that inventory replenishment is done periodically. In order to manage procurement, your executive team will first have to set inventory targets for each product. Through the MRP calculation, the system will then calculate the quantities that must be purchased to meet those desired levels of stock. Say, you have 345 units in inventory and that your inventory target is set at 500 units, the MRP calculation will create a requisition for 155 units (500 minus 345). Then, a member of your team will have to convert the requisition into a purchase order (PO). The PO will be automatically sent to the pre-assigned vendor.

Two key decisions must be made by your executive team: (i) what are the inventories targets set for each product, and (ii) how often and when MRP should be run and purchase orders should be sent to the vendor.

Three key aspects must be considered in order to set your replenishment strategy:

1) The company has one central warehouse and three regional storage locations. Together, the main warehouse (storage location 03) and the three regional storage locations (storage locations 03N, 03S and 03W) have enough room for a total of 4000 units of products. If your company requires more warehouse space, you can rent additional space on a daily basis. The additional space can be rented at a cost of: €300/day for each 1 000 additional units. The cost of additional storage will automatically be billed and paid immediately. Hence, if you prefer to avoid paying storage fees, your total inventory targets for all products should NOT exceed 4 000 units.

WAREHOUSING COSTS	
Current Space (Units) (maximum capacity without additional fees)	4 000
Daily Cost per Additional 1 000 Units	€300

Warehousing Cost for the Extended Game

NOTE: you might be able to offset the higher storage fees by avoiding PO fees (transport fees that your supplier charges with each transfer). In other words, ordering in higher volumes means that you pay less transfer fees, but you will pay more storage fees.

2) Whenever your company sends a purchase order, it will take 1-2 (virtual) days before you receive the material in the central warehouse. It will take one extra day to ship it to the desired regional storage locations. Hence, it takes between 2-3 days before any product ordered can be sold to your customers. That delay should be taken into account.

3) Finally, whenever your company sends a purchase order to the vendor, transportation fees of €1 000 are charged by your suppliers.

If you choose to procure a lot of units in advance in order to meet market demand, you might have to pay the additional storage fees. If you choose to procure only small amounts of boxes in advance, you may either run out of stock and lose sales or have to reorder more often and incur additional transportation fees. Hence, there is trade-off between ordering too much and ordering too little and too often. Your team must use market data, in order to set the proper inventory targets and replenish in a timely fashion.

THE SIMULATION AND ASSOCIATED RULES

In this section, we will provide several important tips to help you as you start to learn the simulation game. While we've attempted to embed the game in a rich context and scenario and make it sufficiently challenging and fast-paced so that you will not be bored, it is nonetheless a simulation game, and therefore a vast simplification of reality. There are some important characteristics that are specific to the simulation which may not immediately resonate with your perception of what can or can't be done in real life.

The rules proposed below have two main objectives. The first and most important one is fairness. The second objective is to make the game challenging and instructive without being overly complicated.

First, we wish to introduce you to some of the important detail and rules of the game.

Game Duration

The game is divided in rounds of about 10-15 minutes. The game can last up to 8 rounds of 10 virtual days each. The simulator is set to perform a virtual day in approximately 1½ minutes. Your instructor can even reduce a virtual day to 1 minute. The game moves fast. Get ready!

End of Game Inventory

In the last round, it is not recommended to sell under cost to clear your inventory. At the end of the game, your unsold product will be valued at the purchasing cost in your profit and loss statement. We encourage you to develop a long-term vision for your firm. On the final day, inventory levels should be able to respond to market demand as if the simulation were to continue a few more days.

Analyzing Market Data

Consumers do not change throughout the game, hence you may learn from their past behaviour.

At the beginning of the simulation, the simulator creates a large population of consumers, each assigned with a "preferred product". Customers do not change throughout the game; hence you may learn from their past behavior in order to forecast regional and product demand. As the game progresses, you should figure out the regional "preferences" by using your transactional data and adapt your plans accordingly.

A Competitive Game

The company with the highest company value at the end of the simulation wins the game.

We compare the performance of the teams by simply using the company valuation of each company at the end. Company valuation is determined by profit and credit rating.

Ethics

You must behave ethically at all times.

The simulation game is fun when it is played competitively. As in real-life, cheating may prove to be an easy way to get a head start, but it should never be an option. We ask participants of the simulation game to act as they would in real life, with the highest ethical standards. Cheating in this game includes the use of any transaction not explicitly permitted in this guide, or by your instructor, and that is used for the sole purpose of increasing the firm's profit or credit rating, or to bypass time delays such as those with supplier delivery.

There are several ways to cheat in this game but be advised that the ERP system and the simulation software keeps track of all transactions. Your instructor is highly likely to call out those who feel the need to play unfairly!

ELEMENTS OF A WINNING STRATEGY

Do Not Run Out of Stock

One of the most important tips for the game is "do not run out of stock". Even if you have the best sales strategy, if you do not have products to sell, it will not matter.

There are 2 actions you can take to avoid running out of stock: 1) reducing demand by increasing the price (short term solution), and 2) increasing the supply by increasing the forecast for that product (long term solution). Managing inventories of popular brands requires operational excellence and is a key element in a winning strategy.

Delays and Capacities are Important Elements of the Game

Delays and warehousing capacities are built into the simulation. These features are important because they limit your ability to react quickly to changes in the marketplace.

To have a product available for the customer, you will have to account for the delay in delivery of the products. This means that you will need to plan in advance and maintain the inventories of products to sustain your sales strategy. However, this will require some liquidity. You must be careful not to buy too much products or hold too much inventory.

The challenge is to be able to find the right balance between low inventory levels and readiness to respond to market demand.

Exploration and Exploitation

To begin the game with a precise strategy is a little presumptuous. You must accept that the simulated marketplace is full of surprises. Learning and adapting should be your motto. Use your data to learn and use the system to adapt to the market.

Most members of your team have the misfortune of having little experience. Fortunately for you, it is the same for all your competitors. To fully exploit the potential of your business, you will need to learn more about the market and industry. You should use a strategy of “exploration and exploitation”.

The beginning of the game should be devoted to exploration. You should experiment with different pricing strategies, transfer amounts, and procurement levels, with the objective of learning more about the market and what works and what does not.

Once you have identified a profitable niche (or niches), you should design your strategy around these niches and exploit and defend them as much as possible.

Organization Matters

A large number of tasks need to be coordinated. Your team needs to (i) keep track of market and sales information, (ii) follow the evolution of stock and cash, (iii) select the allocation level for storage locations, (iv) procure products in a timely fashion, (v) set sales prices, etc. Good teamwork is essential. If your team members work in silos, you will undoubtedly fail. You cannot do this if the roles and the information flow within the team are not well adapted to the requirements of this fast-paced game. You face organization challenges and you must learn how to delegate roles within your team. There are many possibilities for innovation that will make your team more efficient; innovation may ultimately make the difference.

Have Fun!

This is only a game! So, enjoy being a manager using a real-life ERP system. Ultimately, this is about learning not winning.

The business simulation presented in this document is an obvious simplification of reality. There are numerous elements of running a business that have not been included. Yet the game remains a complex affair. You must be able to extract the information necessary to make timely decisions from a large set of transactional data. You need to operate this virtual enterprise in an accelerated time frame. We have taken the point of view that if there is no challenge then there is no fun, and if the simulation is no fun and unchallenging then one does not learn as much. You are asked to run a company at an accelerated pace with a real-life highly complex ERP system. You are bound at one point or another in the game to make errors and find yourself in trouble. Cursing the computer generally will not help. Ask your friends or the instructor for help and keep smiling.

Ultimately, what matters is what you learn from the experience – we hope you learn a lot.



PART 2

The ERP System

ERP software is an integrated information system that manages enterprise data, helps integrate business processes, and provides data for business intelligence. One of the first things to learn is how to perform the various operational tasks to plan procurement, order materials, enter prices, choose the allocation logistics, etc. As the system stores all relevant information about your company, you will also learn how to use the system to exploit that information. You will need to track purchasing orders and inventory levels, discover market preferences based on sales reports and monitor the health of your company by drawing financial reports.

By using SAP to play the game, you will quickly have to:

- Become familiar with the Enterprise Resource Planning (ERP) software
- Develop a practical understanding of the main concepts underlying an ERP software
- Identify the benefits of intra-company integration

Each company in the game is using an integrated information system (ERP) to manage its operations. The idea is to use all existing information to avoid data re-entry, and to create and store new data for future use in other transactions.

An enterprise system is built around an integrated database. A system like SAP contains tens of thousands of data tables. When a user creates, changes or views data in the system, he or she performs a transaction. Data changed or created by one user can be exploited by all other users (e.g. reporting, tracking, execution of orders, etc.). This is the virtue of an integrated system.

A large number of pre-programmed transactions exist in SAP. All transactions are traceable; for legal and controlling purposes, transactions with financial impacts can be reversed but not erased. Each transaction has a transaction code or technical name – a user can execute a transaction if one knows the transaction code. An alternative way to select a transaction is to use the SAP menu. One can browse the SAP menu by drilling down through the successive folders and finding the desired transaction.

The standard SAP menu provides access to a very large number of transactions. For the purpose of the ERP Simulation Game, we have created a user menu that contains only the transactions mentioned in this guide. For each transaction that you will use, we provide both the transaction code and the path in the user menu to access it. Many transactions take you directly to the screen to view, edit or create information. Others require you to first fill in a selection screen, where you enter criteria to filter the information you will be using.

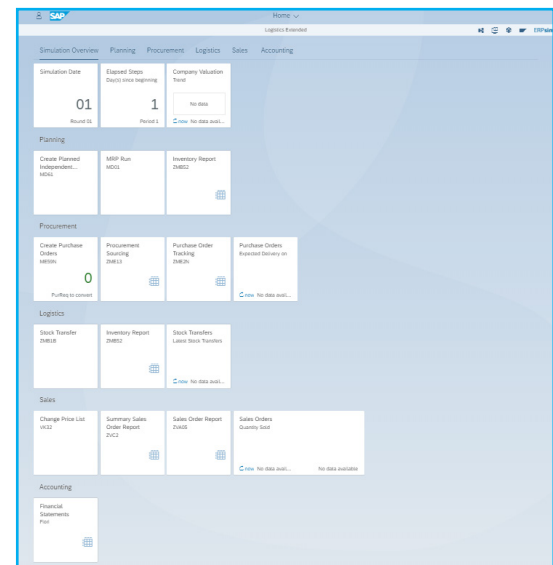
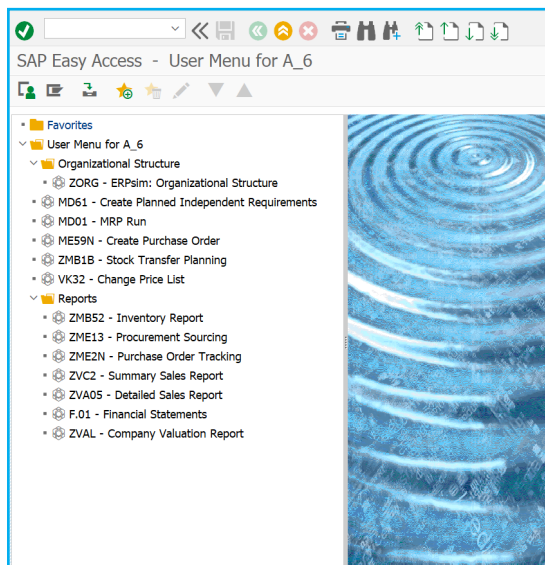
Before moving on to the specific transactions used for ERPsim, we will cover a few navigation basics.

CHAPTER 3

Getting Around in the SAP System

USER MENU

After you log in to the SAP system, you will be presented with a user menu much like the one that can be seen in the figure below. This menu is customized to the logistics simulation, and has some variations depending on the scenario: Introduction, Extended. In all cases, transactions are only added as you progress through the scenarios, never moved or removed. All the transactions are grouped below the user menu (see the figures below). Once you learn where something is, you will always be able to find it. All the reports are grouped together into their own top-level folder for convenience.



*User Menu for the Logistics Extended Scenario
Left: SAP GUI Menu and Right: Fiori Launchpad Menu*

To run a transaction, you can double click on it from the menu or use its technical name (code).

Despite the ease with which you can run a transaction if you know the technical name, SAP will not allow you to run a new transaction while you still have another transaction open. You must first return to the user menu before you can run another transaction.



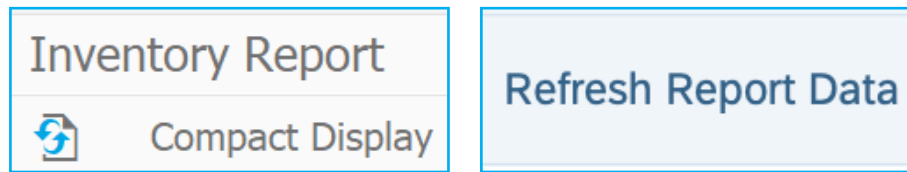
TOOLBARS AND OPTIONS

At the first glance, the structure of buttons and controls may not be obvious, so let's go through a quick orientation. The very first row of buttons is the main toolbar, shared by all transactions, and available from the user menu even when no transaction is open.

Note that not all the buttons are available, depending on which functions are supported by the open transaction at any given moment. The most useful buttons for you to know are as following (Fiori and GUI buttons respectively):

Continue		Enter	Confirm entry of information and/or progress to next screen
Save		Save	Save your information
 Exit		Navigation	Back (one screen or step), Exit (back to user menu), Cancel (Abandon changes and exit). The behavior of these sometimes varies depending on the transaction. Sometimes Cancel will not exit a transaction, and Exit will take you back one screen rather than exiting all together!
N/A*		New session	Opens a new window. You can open up to 6 windows. *In the Fiori Launchpad, clicking on a tile from the User Menu opens up a new tab page in your browser.
		Command box / Transaction launcher	If you know the technical name (code) for a report you can type it here, then press the enter key, to run it.

Below the main toolbar, is the transaction header and transaction toolbar. This toolbar is specific to the open transaction.



On many reports you will see a Refresh icon ( / Refresh). Use this button to refresh the data displayed in the report.

Finally, on some transactions, typically reports, you can see a data grid that also has its own toolbars and headers.

Inventory Report: Round 1 Day 01				
				
SLoc	Material	Description	Stock	
03	AA-T01	Milk	0	
	AA-T02	Cream	0	

Many custom reports are available that look like this, and conveniently have a line just after the toolbar and before the data that tells you the current day in the simulation. The toolbar has options that let you sort, total, subtotal and filter the data in the grid.

SELECTION SCREENS

In many transactions you will first see a selection screen, with many criteria that you can use to narrow the scope of the data that you will be maintaining. Many of these have been configured with default values to help you, but there are a few you will have to fill in for yourself. Some fields require only a single value, others allow a range to be specified. In the example shown in the figure below, the value 16 is specified for Distribution Channel.

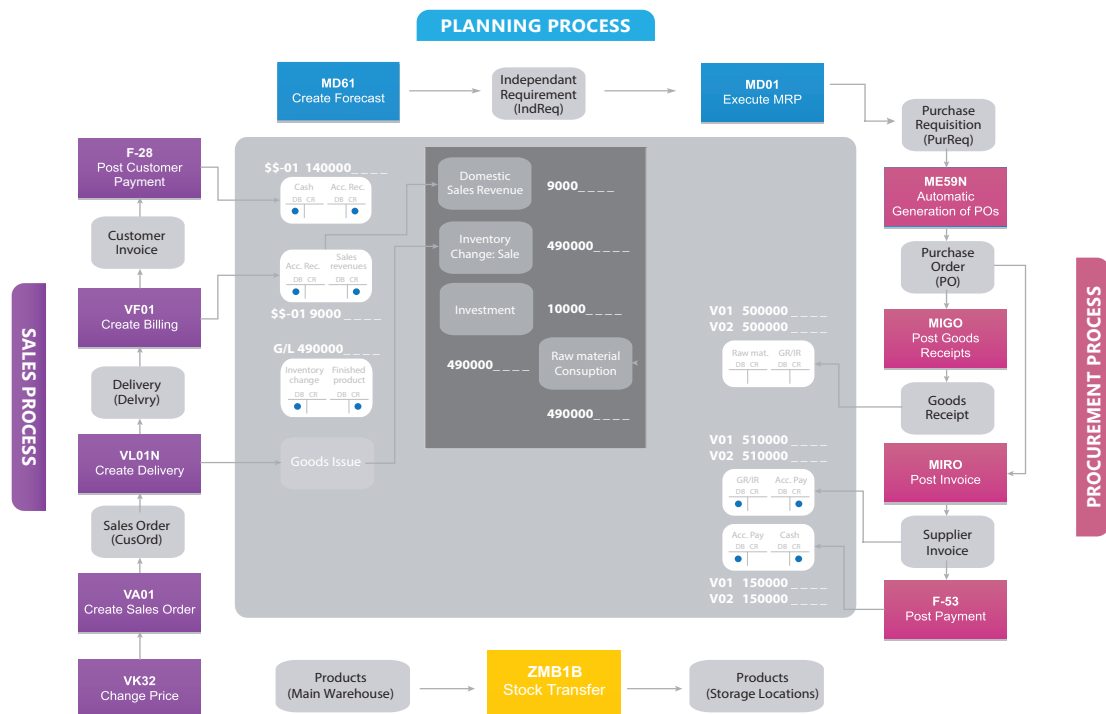
After entering values in the selection screen, you click on Execute ( / Execute) in the transaction specific toolbar to submit your selections and retrieve or process data.

The image shows a SAP selection screen titled 'Price List with Release Status'. It includes a toolbar with 'Save as Variant...', 'Get Variant...', 'Dynamic selections', 'More', and 'Exit'. The main area contains input fields for 'Sales Organization' (A6), 'Distribution Channel' (16), and 'Price List Type'. There are also 'to:' fields for range selection.

CHAPTER 4

Transactions Used During the Simulation

In this section, we cover in detail all the transactions used during the simulation. As mentioned previously, the simulation automates many transactions – those transactions, which are not used during the simulation, are covered in the cash-to-cash exercise document. For reference, figure below shows the complete business process and associated transactions. We will present the transactions in progression order, starting from Logistics Introduction Game and advancing into the extended game. 'Report' transactions (e.g., sales report), which are shared among many functions, will be discussed with the business functions where they are first encountered.



LOGISTICS

ZMB1B

Logistics / ZMB1B –
Stock Transfer
Planning

Relevant Scenarios:

Introduction
Extended
Platinum - Logistics
Manager

Stock Transfer Planning

The stock transfer planning transaction is where you maintain your shipping plans, deciding how much product to send, and how often, from your main warehouse to your regional warehouses.

1. Select your Planning Mode: either Pull or Push. We recommend that you start with the push allocation rule. You can try the other rule later.
2. Enter your Scheduling frequency in days.
3. In the Allocation Plan grid, enter the quantity for each product and region. Remember, with a push strategy, these are the amount of product to ship each time a truck is sent. For a pull strategy these are the target inventory levels to maintain for each product and region, with the amount to put in the truck calculated each time a truck is sent.
4. Press the Enter key. Under and to the right of the grid, the subtotals will update. Verify that these subtotals by product and region are in line with your overall strategy.
5. Click Save ( / **Save**). A confirmation message appears.

OTHER INFORMATION AND OPTIONS

When you enter in this transaction, you will see the current stock transfer plan. You will continue to follow this plan of stock transfer amount and frequency until you change it.

To stop sending products to the regions, or to more quickly change your whole plan, you can click on 'Clear' ( Clear) to reset all the values in the grid to zero. Clearing does NOT save. After clearing you must click on Save ( / **Save**) to record your decision.

SALES AND MARKETING

VK32

Sales and Marketing /
VK32 – Change Price
List

Relevant Scenarios:

Introduction
Extended
Platinum - Pricing
Manager

Change Price List

In SAP, a price list is just one part of many conditions that can determine final pricing when a sales order is created. The pricing procedure looks for condition records to set the prices based on the various conditions applicable when a product is sold. These condition records include the price list, discounts, surcharges, transportation costs and taxes. In the simulation, we only use price list conditions, so you must specify a sales price condition record for each product that you wish to sell to distribution channel 16. To do so, follow the steps below:

1. After running VK32 – Change Price List you will see the “Change Condition Records” screen, with several folders.
2. Open the “Prices” folder by clicking on the small triangle to the left of it.
3. Double click on the “Price List” item. A selection screen appears.
4. Check that the value for the DC field is 16. If the field is empty or shows another value, enter 16 or change the value to 16.
5. Click Execute ( / **Execute**). The list of products and prices appears.
6. Edit the prices in the Amount column.
7. Click Save ( / **Save**). The screen closes and a confirmation message appears.

OTHER INFORMATION AND OPTIONS

In the selection screen (step 4), you can also optionally enter a material code if you want to change the price for just one material. Otherwise, leave the fields as they are!

Also note that only one person at a time could change the prices. If one of your teammates is already in the price list looking at or changing prices, other team members will be blocked from entering the price list page.

Relevant Scenarios:

Introduction

Extended

Platinum - Stock
Manager

Inventory Report

One of the key elements of the game is to be able to react to customer demand and ensure that the inventory of your products is maintained at the right level. Each time the delivery of products by suppliers is confirmed, the inventory of the product goes up. Each time a sale is processed the inventory goes down accordingly. Monitoring inventory levels will help you forecast demand and decide when you should run MRP and send orders. It also helps you monitor and adjust your regional transfers.

Find and click on Inventory Report from the 'Reports' folder in SAP GUI or under the Logistics or Planning tabs in SAP Fiori. There are no additional steps. Use the refresh button to update the data displayed in the report.

Inventory Report Column Descriptions

SHORT NAME	FULL NAME	DESCRIPTION
SLoc	Storage Location	In the SAP system, a storage location is a logical grouping of materials, used to manage like inventory. In the logistics game there are 4 storage locations: 03: This is for your main warehouse. 03N, 03S, 03W: These are your regional distribution centers in the North, South and West respectively.
Material	Material	This the SAP code for the product.
Description	Material Description	This is the name of the product.
Stock	Stock	This is the quantity on hand of a product (e.g., Milk) in a specific storage location (e.g., North).

OTHER INFORMATION AND OPTIONS

In addition to the material stock listing by location information, a Warehouse Capacity block is also included in this report (on top right-hand side). This block shows the capacity of the main warehouse and regional warehouses (4,000 for introductory and extended games) as well as the total quantity of materials on hand. Effectively, you can see how much of your storage capacity is in use at a glance.

On the toolbar, there is a standard Refresh button, and a "Compact Display" option. This later button moves the position of the Warehouse Capacity information block from the right of the stock material listing to above it. Once clicked, the button will read "Widescreen Display". Click to move the block back to the right of the list.

Summary Sales Report

A fundamental task in your business is monitoring and analyzing sales orders to understand the market demand for your products. The summary sales report gives you an overview of your sales each day, broken down by product. This report is extremely helpful, as it tells you how often each product is being ordered, as well as a feel for the profitability of each product – the revenues less the cost of goods sold. Your best products are the ones that are being ordered often (preferably several times a day, every day) that bring a lot of revenue for little cost. You may be tempted to only look at the quantity sold – quantity is not everything. It is easy to sell a lot of product by lowering your price, but you will not generate a lot of profit. Focus on maximizing revenues and profit, not quantity sold.

Find and click on the report from the 'Reports' folder in SAP GUI or under the Sales tab in SAP Fiori. There are no additional steps. Use the refresh button to update the data displayed.

Summary Sales Order Report Column Descriptions

SHORT NAME	FULL NAME	DESCRIPTION
Rnd	Round	The simulation round in which the sales were recorded.
Day	Day	The simulation day on which the sales were recorded.
Material	Material	The SAP code for the product.
Description	Material Description	The name of the product.
Orders	Orders	The number of orders that included the product, on a given day.
Qty	Quantity	The total number of units of cereal ordered for that product on a given day.
Value	Value	The total revenue earned for that product on a given day.
Cost	Cost	The value of the inventory that was sold for that product on a given day – cost of goods sold.

OTHER INFORMATION AND OPTIONS

The Sales Order Report can be modified to organize the information differently. The report contains a table. The data contained in the table can be sorted or filtered. In order to filter the data, click on the icon . In order to sort the data, first select a column corresponding to the sorting criterion and click on  to sort data in decreasing order or on  to sort in increasing order. Totals and subtotals can also be calculated and displayed. Select a column that contains a numerical field that can be summed – e.g., sales revenues – and then click on the summation icon . This will provide the total sales revenues. In order to obtain subtotals – e.g., sales revenues by round and virtual day – select the round and virtual day columns and click on the subtotal icon . This will generate the sales revenues for every round and for every day.

ZVA05

Sales and Marketing
/ Reports / ZVA05 –
Detailed Sales Report

Relevant Scenarios:

Introduction

Extended

Platinum - VP Sales

Detailed Sales Report

To review the sales orders received by your company in detail, use the detailed sales order report. This report displays every product ordered for every sale. Most importantly, it has information about the 3 geographic regions (North, West, and South). Use the summary report to guide your overall pricing strategy. Also, use this detailed report to find the market preference for each product in each region and fine-tune your transfer schedule to these regions. Comparing to the summary sales report, the detailed sales report has therefore 2 additional items: Area and Price. However, it does not have the item 'order'.

Find and click on the report from the 'Reports' folder in SAP GUI or under the Sales tab in SAP Fiori. There are no additional steps. Use the refresh button to update the data displayed.

Detailed Sales Order Report Column Descriptions

SHORT NAME	FULL NAME	DESCRIPTION
Rnd	Round	The simulation round in which the sales were recorded.
Day	Day	The simulation day on which the sales were recorded.
Area	Area	The geographic region the customer is in – North (NO), South (SO) or West (WE).
Material	Material	The SAP code for the product.
Description	Material Description	The name of the product.
Price	Price	The unit price that the product was sold for.
Qty	Quantity	The number of units sold.
Value	Value	The revenue earned by the sale of that product (Qty X Price)
Cost	Cost	The value of the inventory that was sold – cost of goods sold (Qty X Standard Price).

OTHER INFORMATION AND OPTIONS

Similar to the summary sales report, the detailed sales report can be modified to organize the information differently. To do so, use filters and sorting options as described in the summary sales report section.

PLANNING AND PROCUREMENT

Create Planned Independent Requirements

This is where you enter your sales forecasts, known as planned independent requirements in SAP. The MRP will use these values as target inventory levels, generating a 'purchase requisition' for the shortfall between the target values and the inventory & in-process purchase orders.

1. On the "Created Planned Independent Requirements: Initial Screen" enter the required selection filters. Select Product group option, and enter \$\$-T in the field to the right of this option. (replace \$ with your team letter).
2. Press Enter ( / **Continue**). The "Plnd Ind. Reqmts Create: Planning Table" screen appears. The grid with products (material numbers) and columns with months is where you will enter and maintain your forecast.
3. Enter your forecast values in the **second column**. This second column will correspond to the next month. Since you are forecasting for future sales we use this column instead of the current month. Since the real SAP system date and time will not move forward during the simulation, the future months will never approach, so don't use the other columns either. Enter values into the second column only – do not use the other months.
4. Click Save ( / **Save**). A confirmation message appears, and you are returned back to the "Created Planned Independent Requirements: Initial Screen".

MD01

Planning and
Procurement/ MD01 –
MRP Run

Relevant Scenarios:

Extended
Platinum -
Planning Manager

MRP Run

The MRP – Material Resource Planning - is an automated tool that helps generate planning documents according to the existing requirements and resources. In the context of the logistics game, its calculation process uses the planned independent requirements, available inventory counts, and in-process purchase orders to generate 'Purchase Requisitions' to meet these requirements.

Running the MRP process is simple:

1. On the "MRP Run" initial screen, default values and settings have already been set for you. You do not need to enter anything.
2. Press Enter ( / **Execute**). A warning message "Please check input parameters" appears. For most organizations with thousands of products, components and raw materials, running the MRP can be a lengthy process. Getting it wrong, means setting up a production and procurement plan that you don't want. This is why the system asks you to verify the settings. Don't worry, the correct ones are set for you – just don't change them!
3. Ignore the orange warnings. Press two more times on Enter ( / **Execute**).
4. In the pop-up window, click on Continue ( / **Continue**). The MRP process runs, and a summary of the results appears.

OTHER INFORMATION AND OPTIONS

After running the MRP, there is a section titled "Database Statistics". Here you will see how many purchase requisitions were created by the MRP process.

No requisitions created – If the MRP didn't generate any new purchase requisitions, then there's no need to procure any materials. This means that the inventory and/or the orders sent to the vendors are enough to cover the forecast amount.

ME59N

Planning and
Procurement/ ME59N
– Create Purchase
Order

Relevant Scenarios:
Extended
Platinum -
Planning Manager

Create Purchase Orders

The MRP process creates purchase requisitions, which are internal documents that signal the procurement department to evaluate different sources of supply, and then place the orders with a selected supplier. SAP provides a simple way to accept all purchase requisitions and aggregate them in creating new purchase orders. The transaction code ME59N automatically creates consolidated purchase orders for each vendor.

Prior to doing this, you can still readjust your forecast and re-run the MRP if you change your mind. Once you create and send the purchase orders, you commit to making the purchase. You cannot change your mind after the order is placed.

1. At the “Automatic Creation of Purchase Orders from Requisitions” selection screen, all the required fields are filled by default. You do not need to enter anything.
2. Click Execute ( / **Execute**). The conversion process runs, and a summary of the purchase orders created appears.

OTHER INFORMATION AND OPTIONS

The system will convert the requisitions (if any) into a purchase order. All requisitions preassigned to a vendor will be grouped in one purchase order. While a requisition is a planning document that can be changed or deleted without problems, a purchase order, once sent, is a formal contract between your company and your vendor and it cannot be changed or deleted. You must now wait for the order to be filled by the vendor, and the items to be received into your warehouse.

No suitable requisition found – At some point during the simulation, you are bound to see “no suitable requisition found” on the bottom line of your SAP screen. The system is telling you that there were no new requisitions that needed to be processed. To see this message, execute the procedure again. Since you just created all the purchase orders you needed and have not rerun the MRP, there are no new requisitions that must be processed. If you see this message, you have either already run the process, have forgotten to run the MRP, or the MRP has not generated any requisitions.

This might be the case if you already had sufficient inventory and purchase orders in the system when you ran the MRP. Retrace your steps and ensure you ran all the transactions properly and in the correct sequence.

ZME2N

Planning and
Procurement/
Reports/ ZME2N
– Purchase Order
Tracking

Relevant Scenarios:

Extended
Platinum -
Planning Manager

Purchase Order Tracking

The table below describes the many different columns of information available in this report. Note that these descriptions are written as though all orders are yet to be received, and activity occurs in the future. During the simulation, some purchase orders will have already been delivered, and their dates will be in the past.

Purchase Order Tracking Report Column Descriptions

SHORT NAME	FULL NAME	DESCRIPTION
Order	Order	The unique number assigned by the system to the purchase order. Note that since there are multiple items per order, a “blank” value indicates that this is another item on the same order as the previous one.
Vendor	Vendor	Which supplier the order has been placed with, as identified by their code in the system. A “blank” value indicates this item is part of the same order as the previous one. A purchase order can only ever be for one vendor.
Material Description	Material Description	The name of the item ordered.
Quantity	Quantity Ordered	The amount ordered; in kg for food material, and by item count for packaging.
Price	Price	The purchase cost per price unit.
Price Unit	Price Unit	The amount ordered that is the basis for multiplying by the price to get the total cost. 1kg for food material, and per item for packaging.
Value	Value	The total cost of the material amount ordered (Quantity / Price Unit) * Price
Status	Status	The condition of the order. Unconfirmed – The supplier (simulator) is yet to confirm that the order has been placed. Expected – The supplier has confirmed the order and the materials have not yet been delivered. Delivered – The materials have been delivered to your warehouse.
Goods	Expected Goods Delivery	The simulation date on which the materials will arrive at your warehouse. N.Sch - Indicates that no delivery date has been communicated yet.

ZME13

Planning and
Procurement /
Reports / ZME13 –
Procurement Sourcing

Relevant Scenarios:

Extended
Platinum –
Planning Manager

Procurement Sourcing

This report shows the price for each material by each vendor. This information will help with pricing decision of the finished products.

Procurement Sourcing Report Column Descriptions

NAME	DESCRIPTION
Vendor	Which supplier sells this material, as identified by their code in the system.
Material	The code of the item ordered.
Gross Price	The price set by the vendor for the given material.
Fixed	If there is an x, it means that the vendor has been selected as the supplier of that given product.
MRP	Here, there are three different options: 1: Purchase requisitions that are generate by MRP, are automatically assigned to this source of supply (vendor). 2: If the source of supply is a scheduling agreement, MRP can create a delivery schedule line with respect to a requirement. If the field is blank: The source of supply is not taken into account by MRP.

ACCOUNTING

Financial Statements

In SAP®, you can get current financial statements at any point in time during the simulation to get a snapshot of your financial situation.

Transaction F.01 is used to display your financial statements:

1. In the "Company code" field, enter your company code (To find your company code, refer to transaction ZORG).
2. In GUI, towards the bottom of the screen (you may need to scroll down), under "List In output" choose the ALV Tree Control option. In the Fiori Launchpad, input SIM1 in "financial statement version".
3. Click Execute ( / ).
4. If a warning message appears, dismiss it by clicking on Continue ().
5. The financial statements appear.

This standard report provides you with the balance sheet and the income statement of your company. You can get more details on each section, down to individual account balances, by clicking on the triangle on the left hand side of every item.

The net income line corresponds to the cumulative profit (or loss) of your firm since the beginning of the game.

This report is a standard SAP transaction and has not been enhanced. It does not have a refresh button. To refresh the data, you will need to go back to the selection screen and execute ( / ) again.

OTHER INFORMATION AND OPTIONS

This standard report provides you with the balance sheet and the income statement of your company. You can get more details on each section, down to individual account balances, by clicking on the triangle on the left-hand side of every item.

The net income line corresponds to the cumulative profit (or loss) of your firm since the beginning of the game. Starting the next round, the report will show the financial results at the end of the previous round. The difference column shows changes for the current round, allowing you to get an idea of your current performance against previous rounds

This report is a standard SAP transaction and has not been enhanced. Therefore, it does not have a refresh button. To refresh the data, you will need to go back to the selection screen and execute again.



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AFTERWORD

BY PIERRE-MAJORIQUE LÉGER

When hired as a young professor at HEC Montréal, I was given the responsibility to teach ERP systems to undergraduate and MBA students. I was looking for new ways to teach ERP Systems using SAP® when I attended a training seminar led by Dirk-Jan Schenk and Casper Draijer, organized by SAP Canada in Halifax. These two professors from HES Amsterdam had developed a role playing game and it turned out to be truly inspirational. It convinced me then that this was the way to go.

After returning to Montreal, I spent the summer developing an ERP Simulation game. It was first used in October 2004. While it generated much enthusiasm from the participating students, the simulation game lacked a well developed and realistic economic story line. To address these issues, I invited Professor Jacques Robert, a friend and colleague at HEC Montréal, to participate in the development of this market game. Professor Robert is an economist and a trained game theorist. He developed a very challenging market algorithm to support the game. Since then, he has been instrumental in the development of most aspects of the actual game version.

Building on this market model, a first prototype of the simulation software was developed in Excel. The simulation engine generated text files that were uploaded and executed using a CATT script in SAP®. The simulation software created customer orders per round, but it had two problems. First, it required far too much human intervention to operate the simulation game, and second, orders could only be created in batches once per round.

This is when Professors Gilbert Babin and Robert Pellerin came into play. Professor Babin, also a friend and colleague at HEC Montréal, is a computer scientist specializing in distributed systems. Professor Pellerin is an industrial engineer from École Polytechnique de Montréal and a former doctoral colleague. Thanks to both of these researchers, ERPsim was developed. ERPsim is a Java-based application that supports almost every aspect of the simulation game. Sales orders are now automatically recorded in the system in real time. Both Professors Babin and Pellerin have significantly contributed to the development of the simulation game. Finally, Professor Bret Wagner from Western Michigan University joined our team in December 2006. With his help, a dedicated simulation client was developed to support the simulation game. Our team is a truly interdisciplinary team.

We have also benefited from the assistance and insightful comments of many faculty members of the SAP University Alliances. The ERP Simulation Game is now used by more than 250 faculty members and in more than 100 universities worldwide. We would like to thank all these colleagues for their comments that have helped improve the software and streamline the simulation game.